Agricultural Data Collection Specification

Development-focused specification for collecting agricultural data from the sources identified in the project blueprint.

1. ICAR — Crop Knowledge

Data fields to collect
• Crop name
• Crop variety
• Agro-climatic/location suitability
• Sowing/transplanting period
• Seed/planting material information
• Land preparation
• Nutrient/fertilizer management
• Irrigation requirements
• Weed management
• Pest management
• Disease management
• Harvesting
• Post-harvest management
• Recommended agricultural technologies
• Source institute / ICAR centre
• Source document/page
• Publication/update date

2. KVK — Local/Location-Specific Advisory

Data fields to collect
• State
• District
• KVK name
• KVK address/location
• Contact information
• Crops covered
• Current crop advisories
• Pest/disease advisories
• Recommended farming practices

• Farmer training information
• Demonstration/technology information
• Advisory date
• Advisory language
• Advisory source URL/document

3. IMD — Weather + Agromet

Data fields to collect
• State
• District
• Block/location
• Forecast date
• Maximum temperature
• Minimum temperature
• Rainfall
• Relative humidity
• Wind speed
• Wind direction
• Cloud cover
• Weather warnings
• Crop-specific agromet advisory
• Advisory date
• Crop/crop stage if available
• Recommended action

4. AGMARKNET — Mandi/Market Data

Data fields to collect
• Date
• State
• District
• Market/APMC
• Commodity
• Variety
• Commodity group

• Arrivals
• Minimum price
• Maximum price
• Modal price
• Unit
• Market/source

5. Soil Health Card — Soil Data

Data fields to collect
• Farmer/farm reference
• Location
• Soil sample date
• pH
• EC
• Organic Carbon
• Nitrogen (N)
• Phosphorus (P)
• Potassium (K)
• Sulphur (S)
• Zinc (Zn)
• Iron (Fe)
• Copper (Cu)
• Manganese (Mn)
• Boron (B)
• Fertilizer recommendation
• Soil amendment recommendation
• Testing laboratory/source

Example Market Record

```
date: 2026-10-28
state: Odisha
district: Khordha
market: Khordha APMC
commodity: Paddy
variety: ...
arrivals: ...
```

```
min_price: ...
max_price: ...
modal_price: ...
```

Recommended Backend Structure

Table	Core fields
FARMER	location, district, state, language, crops
CROP_KNOWLEDGE	crop, variety, growth_stage, practice, pest, disease, fertilizer, irrigation, source
KVK_ADVISORY	district, kvk, crop, advisory, date, source
WEATHER	location, date, temperature, rainfall, humidity, wind, warning
MARKET_PRICE	date, market, commodity, variety, arrivals, min_price, max_price, modal_price
SOIL	farmer/farm, pH, EC, OC, N/P/K, S, Zn/Fe/Cu/Mn/B, recommendations

Development Principle

Location → Weather → Crop → Crop Advisory → Soil → Market → AI response.

Collect only the structured agricultural knowledge needed for these product features. Store the source document/page or URL alongside each knowledge/advisory record so the AI layer can trace the origin of an advisory.